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The Text Beside the Image: Detection, Utility and Leakage for Trustworthy Multimodal Medical Data and Beyond

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Abstract. Medical images are released with the free text that describes them, and protecting the image does not protect the report. We measure identifier detection, downstream utility and residual identity leakage on the same documents, with the pseudonymisation policy as the variable under test: 15 detectors, three release conditions and four corpora of clinical letters, legal judgments, news and other genres, and e-mail, in German, English, Chinese and Arabic. A fixed 13-detector union replaces almost every person mention in the clinical letters at specificity 0.869, and 0.890 of them in e-mail at specificity 0.624. With this ensemble, frequency matching with a public name list recovers zero identities across the four corpora; its returned names were missed by the detector and printed in clear text. Cross-document linkage reaches 0.0095 without training and 0.0407 with it on e-mail, against 1/3816 chance and a 0.706 ceiling on unmodified text. The tested replacement forms expose the same reachable text to a scheme-aware adversary: the typed-placeholder release is a character-for-character rewrite of the surrogate release. Under this ensemble, residual exposure is set by identifiers the detector missed and must be reported per document, per case and per language.

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Keywords: pseudonymisation · clinical text · multimodal medical data · privacy evaluation

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1 Introduction

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Clinical data are multimodal: a chest radiograph travels with its report, a cardiology study with its discharge letter. Releasing either half requires both to be protected, since an image may need defacing or removal of burned-in text while its report needs identifier replacement. Artificial intelligence scales both tasks, and its errors vary by distribution, language, script and identifier class. Packhäuser et al. [18] re-identified patients from de-identified chest radiographs and priced a defence against diagnostic utility [19]; pathological speech has been evaluated against privacy, utility and fairness [23]; clinical text is instead kept with the model inside the hospital [9]. We ask what survives transformation of the text, and what protecting it costs.

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De-identification reduces identifying information. Reversible *pseudonymisation* replaces an identifier with a realistic *surrogate*; *anonymisation* prevents singling out. Stable pseudonyms preserve longitudinal utility but hand an adversary a persistent handle. European guidance treats keyed message authentication codes as robust [8], and the health-informatics standard is revised accordingly [6].

Detection benchmarks ask whether an identifying span was found [10, 25, 27]; utility studies measure downstream change [1, 26] or information loss [24]; re-identification studies measure attack success. Frameworks combine technical performance, information loss and an intruder test [12], while surrogate systems and masking benchmarks cover single pieces [7, 17, 21]. None states together what one policy buys and what it costs. Data-protection officers release documents and cases, not average tokens, and one unchanged mention can identify someone after thousands were removed, so we measure surviving findings, exposed cases and linkable entities rather than token rates alone.

One design covers four corpora in German, English, Chinese and Arabic: clinical letters, legal judgments, news and other genres, and e-mail. Every document receives the same detector comparison, release conditions and measurements. Two predictions are put at risk: that detector precision governs utility more strongly than the choice between surrogates and typed placeholders, and that detection sensitivity moves the two attacks in opposite directions, because removing a name obstructs context linkage while leaving a larger, more stable population for frequency matching. The clinical corpus has a known weakness: its gold names were inserted into an already de-identified release, so its name distribution is synthetic.

2 Material and Methods

2.1 Corpora

Reference provenance differs, so detection denominators are not interchangeable (Table 1). TAB manually annotates identifiers and document-scoped co-reference in court judgments [21]; OntoNotes covers three languages but was annotated for linguistics rather than privacy. A *named entity* is a predefined person, organisation, location or other type, its occurrences *mentions* forming a co-reference chain. Enron derives genuine cross-document identity from message headers but contains exposed personal information [16]. CARDIO:DE is the clinical anchor, its gold identifiers inserted into an already de-identified release, which makes its name distribution synthetic.

One denominator follows. CARDIO:DE gold covers only inserted identifiers, and 84.7% of its gold tokens are dates, which the detectors do find but the conditions pass through by design. Sensitivity, replacement, exposure and the attacks are therefore all scored over the types actually replaced, and the retained types are in none of those denominators. Retained dates are themselves a re-identification channel, and a release keeping them needs date shifting. A TAB applicant is usually named once but has a median of 17 later role mentions, and only Enron has natural cross-document identity.

Table 1: The corpora separate domain, language, and where the reference annotation came from.

Corpus	Docs	Tokens	Ment./doc	Lang.	Reference annotation
CARDIO:DE	400	885 059	78.3	de	constructed
TAB	1268	1 844 804	81.2	en	manual
OntoNotes	5994	4 726 594	45.5	en/zh/ar	linguistic
Enron	58 636	15 507 925	14.4	en	header-derived

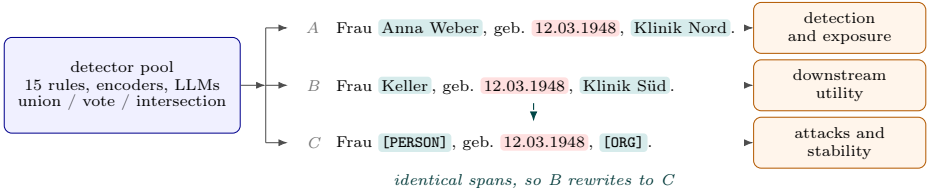


Fig. 1: One set of detector spans generates three release conditions, on an invented header line in the style of a discharge letter. Detected identifiers (teal) become surrogates in B and typed placeholders in C; dates (red) pass through by design. A surrogate replaces a whole detected span with a single name, so the two-token name becomes one. B and C replace the same spans, so C is a rewrite of B. Every condition meets every measurement family on the same documents.

2.2 Conditions

Condition A is unmodified text. In B, each detected identifier becomes a realistic *surrogate*, such as another name, stable across the corpus, indexed by HMAC-SHA256, a keyed one-way digest irreproducible without the secret. Condition C uses an unindexed placeholder such as [PERSON]. B and C differ only in replacement form, following European guidance and prior e-mail pseudonymisation [7, 8], and replace identical spans.

Before hashing, the key normaliser casefolds mentions, collapses whitespace and removes titles and punctuation, so *Dr. Weber* matches *weber* but *Weber* does not match *Kay Weber*. Unrelated people may therefore share a key while one person’s initials and full name may not; Sect. 3.3 reports both failures. Figure 1 evaluates every condition on identical documents and measurements.

2.3 Detectors, Ensembles and Operating Points

The 15 frozen detectors comprise rules, zero-shot and fine-tuned encoders, multi-lingual and German clinical recognisers, and eight gateway language models; the repository lists every model, version and prompt. Cached spans are combined by union, intersection or at-least- k vote. Two rules recur below: the *permissive* rule is the union of all 15, the *precise* rule a majority, 8 of the 15. Leakage uses the configuration we recommend, a fixed 13-detector union; two large language models remain available singly but are excluded from it, because preliminary

runs made them one to two orders of magnitude slower per document without improving detection. A proportional output limit truncated 92 % of Chinese replies, which we corrected and re-ran rather than score truncation as a detection failure.

Each corpus receives fast and maximum points for sensitivity, the share of replaced-type identifier tokens detected, and specificity, the share of all other tokens left untouched; an ensemble run in parallel costs its slowest member. Neither rate is readable alone, so both are given at every point. A maximum point is non-dominated, the fast point the cheapest within 10 % of it, subject to a sensitivity floor that stops an empty detector winning on specificity. This search covers singles, pairs and triples, so the recommended 13-detector ensemble lies outside it and may exceed the maximum of Table 2.

2.4 Measurements

Detection is scored by token and by entity, an entity counting as protected only when every one of its mentions was replaced. Information-weighted precision scales each masked token by how poorly its context predicts it, so removing a predictable word costs less than removing an informative one, and exposure uses document, case and entity denominators, because few leaked tokens can expose many people. Frozen models analyse all three conditions for named-entity agreement, medication extraction and section classification; agreement is span F1 between the outputs on original and transformed text, and the original-text score accompanies every comparison. Continuous pairs use the Wilcoxon signed-rank test, binary correctness McNemar’s test, and Benjamini–Hochberg correction applies within task families. Utility and attack figures carry one standard deviation over documents, detector combinations or folds; detection and exposure are exact population proportions.

A *collision* gives distinct people one surrogate, *fragmentation* gives one person several within a document, and *drift* changes it between documents. These apply to B, which has a mapping, and not to placeholders, which merge every entity of one type by design.

2.5 The Attacks

Frequency matching aligns surrogate frequencies with real-name frequencies. Published attacks on pseudonymised free text target the identifiers a system missed rather than the surrogate distribution it produced, as the parrot attack does [4], so we transfer the alignment model from attacks on encoded identifiers in record linkage and disclosure-risk work [14, 29]. Two priors run over one observation, following Kuzu et al., Niedermeyer et al. and Vatsalan et al. [11, 15, 28]: a corpus-internal prior granting perfect knowledge, an oracle upper bound, and a public name-frequency list, the realistic adversary.

The attacker tags the released text itself and so also sees names the defender missed. Every detected surface stays in the denominator, including ambiguous and unresolved bins [5], and if k entities share a surrogate a recovery scores

Table 2: Operating points on both error rates. Sensitivity is over the identifier types the conditions replace, specificity over every other token. Cost is seconds per document; the rule is $\cup$ union, $\cap$ intersection, v at-least- k vote, s a single detector.

Operating point	CARDIO:DE			TAB			OntoNotes			Enron		
	Sens.	Spec.	Cost	Sens.	Spec.	Cost	Sens.	Spec.	Cost	Sens.	Spec.	Cost
max. sensitivity	0.992	0.9670	57.633 ^u	0.925	0.8986	14.210 ^u	0.747	0.9763	9.450 ^u	0.973	0.7720	25.557 ^u
fast sensitivity	0.895	0.9769	0.301 ^u	0.893	0.9242	0.219 ^u	0.684	0.9731	5.376 ^u	0.899	0.7419	0.057 ^u
max. specificity	0.523	0.9997	116.875 ^v	0.501	0.9959	98.110 ^v	0.503	0.9960	0.103 ^u	0.588	0.9428	20.931 ^u
fast specificity	0.781	0.9862	0.263 ^s	0.520	0.9934	0.117 ^s	0.503	0.9960	0.103 ^u	0.674	0.8787	0.026 ^s

1/ k rather than being deleted. Abstention occurs where $(\max - \max_2)/\sigma < \varphi$, computed from the attacker’s own scores [13]. We report the answered-query rate and $c@1$, which credits warranted abstention [20], beside a chance rate from the candidate set [22]. Realistic surrogates exist so residual identifiers are hard to pick out [3], which is what the public prior tests; sweeping its size and vintage measures what better knowledge is worth [2].

Context linkage ranks identities by cosine similarity between surrounding-word profiles; *learned linkage* fits the same task on labelled, entity-disjoint examples. Both use identical document-disjoint folds, and only the labels of held-out queries are hidden: those people stay in the complete candidate list with every distractor, so neither attack is told that a query has a match, and their difference measures learning rather than gallery size. Cross-document identity permits this on Enron and CARDIO:DE but not on TAB or OntoNotes.

LLM candidate ranking gives a language model one marked mention and ten candidate names, with or without auxiliary text about each candidate from the released condition. Because B and C replace identical spans, B can be rewritten into C, so we report this attack on B as presented, on B after that rewrite, and against a ranker told the scheme. The one-in-ten closed world measures an attacker holding a shortlist, not population re-identification. A *dictionary lookup* would hash candidate names; it is defined but not run, because no condition here is unkeyed.

3 Results

3.1 Detection

Table 2 gives the cost–quality front, with sensitivity the primary criterion for protection. Maximum-sensitivity configurations reach 0.992 on CARDIO, 0.925 on TAB, 0.747 on OntoNotes, and 0.973 on Enron. The faster alternatives retain 0.895, 0.893, 0.684, and 0.899 sensitivity, respectively, while sharply reducing cost; for example, CARDIO falls from 57.633s to 0.301s and Enron from 25.557s to 0.057s per document. The OntoNotes specificity rows coincide because the 10 % band admits all 414 candidates.

Within OntoNotes, language effects depend on what is detected. Under the recommended ensemble, sensitivity is 0.989 on English, 0.868 on Chinese and

Table 3: Leakage from the recommended 13-detector union; all columns concern people. *Sens.* is person sensitivity and *Spec.* the share of non-identifier tokens left alone: neither is readable without the other. *Exp.* is the share of people keeping a mention in clear text, *Frequency* the identities named from frequency alone with a public list and, bracketed, with the corpus’s own distribution (an upper bound, not a risk). *Linkage* is Rank-1 over the complete candidate list against chance 1/gallery.

Corpus	Sens.	Spec.	Exp. (%)	Frequency	Context link.	Learned link.	Chance
CARDIO:DE	1.000	0.869	0.1	0 (1)	0.0000 ± 0.0000	0.0071 ± 0.0160	1/207
Enron	0.890	0.624	18.0	0 (0)	0.0095 ± 0.0012	0.0407 ± 0.0029	1/3816
TAB	0.996	0.850	0.8	0 (1)	not measurable		–
OntoNotes	0.935	0.932	4.9	0 (1)	not measurable		–

0.614 on Arabic; pooled sensitivity is 0.801. Arabic supplies 45 % of identifier tokens from 7 % of documents, so it dominates the pooled result while remaining the hardest case. Person sensitivity transfers better at 0.991, 0.932 and 0.878, pooling to 0.935. The thirty-seven-point spread therefore lies in the other identifier classes rather than the people targeted by the attacks.

3.2 Utility

At the high-sensitivity operating point, utility depends strongly on replacement form. On CARDIO, named-entity agreement is 0.611 ± 0.082 with surrogates but only 0.004 ± 0.008 with placeholders. The latter largely reflects the frozen recognizer rather than semantic degradation: it never tags [PERSON] as a name, so every replaced mention is counted as a disagreement. Surrogates therefore preserve token-level named-entity utility substantially better than placeholders.

The other two tasks retain their original-text baselines and support paired comparisons. Medication extraction scores 0.371 ± 0.182 on unmodified text. Under the precise rule, surrogate replacement yields 0.364 ± 0.189 , with a paired difference of -0.008 ± 0.178 that is consistent with zero ($q = 0.80$). Under the permissive rule, performance drops to 0.346 ± 0.181 , corresponding to a paired difference of -0.020 ± 0.182 ($q = 5.6 \times 10^{-3}$). Section classification scores 0.749 ± 0.093 on unmodified text. Surrogates reduce this score by 0.072 ± 0.086 , whereas placeholders reduce it by 0.038 ± 0.089 ; both effects are significant ($q = 1.6 \times 10^{-37}$ and $q = 2.6 \times 10^{-14}$; rank-biserial correlations -0.846 and -0.531). Thus, typed placeholders preserve more utility at the document level, whereas surrogates are required to preserve token-level utility.

3.3 Leakage

Table 3 reports the 13-detector union. This frequency attack with the public name list recovers zero identities across the four corpora. Its returned names were missed by the detector and printed in clear text, so it measures defender sensitivity rather than adversarial inference. The corpus distributions, unavailable to a real attacker, align at most one identity. Raising the abstention threshold

from $\varphi = 0$ to $\varphi = 0.05$ leaves 11 of 25,677 CARDIO:DE queries and zero elsewhere; alignment $c@1$ is zero at every threshold.

Linkage requires cross-document identity and must be read against unmodified text and chance. On Enron, fixed similarity recovers 0.706 of held-out people from unmodified text and 0.0095 ± 0.0012 after the recommended release. The latter is 36 times the $1/3816$ chance rate but two orders of magnitude below the unmodified-text rate. Entity-disjoint training raises recovery to 0.0407 ± 0.0029 , four times the fixed attack and 155 times chance, so a trained adversary carries the residual risk. CARDIO:DE has 207 attacker’s reference identities; fixed linkage recovery is zero, and learned linkage 0.0071 ± 0.0160 against chance 0.0048 is consistent with guessing. Across the tested Enron configurations, linkage rises as person sensitivity falls. Agreement between two detectors lowers sensitivity to 0.732, raises fixed linkage fivefold to 0.0500 and increases clear-text names from 180 to 1,868; intersection yields 0.037 sensitivity and 0.2380 linkage.

The third attack assigns different risks to the replacement forms. Ranking one marked mention against ten candidates, the model falls from 0.995 on unmodified TAB to 0.110 with surrogates, near 0.100 chance, but reaches 0.340 with typed placeholders; OntoNotes (0.290 versus 0.630) orders likewise. Only CARDIO:DE (0.085 versus 0.120) stays at chance level for both categories. This is consistent with source-text recall, not stronger surrogate protection: a neutral placeholder permits retrieval of a memorized person, whereas a surrogate supplies a conflicting name and suppresses that candidate. Accordingly, C is highest on widely replicated OntoNotes, intermediate on public TAB, and at chance on restricted CARDIO:DE. Yet both forms replace identical spans. Replacing each surrogate with [PERSON] reproduces C character for character on all 1,268 TAB and 5,994 OntoNotes documents.

Sensitivity alone omits exposure units; each denominator answers a different question. The 13 detector union leaves one CARDIO:DE person mention of 4,396 in clear text, one entity of 1,957 and one patient of 270; over every identifier type, 8.5 % of patient letters carry something and 12.2 % of patients do. TAB leaves a name in 44 of 1,268 judgments and OntoNotes in 736 of 5,994 documents, while Enron leaves one in 18.0 % of its people and something in 45.8 % of its messages.

The key normaliser both merges and splits identities: on Enron, ORGANISATION surrogates collide at 17.3 % and drift at 48.6 %, and PERSON drift reaches 37.7 % against 5.0 % on CARDIO:DE. The lower CARDIO:DE figure reflects constructed recurrence with fewer natural variants rather than generalisation.

4 Discussion

For releases prioritising leakage reduction, the measurements support the large union, whose CARDIO:DE specificity is 0.869. Requiring detector agreement trades sensitivity for specificity. On Enron, two votes lower person sensitivity from 0.890 to 0.732 and raise fixed linkage fivefold. Across these tested configurations, lower sensitivity coincides with greater name exposure and linkage.

Attack results depend on adversarial knowledge and denominator. The public prior measures this frequency attack with one available source; its contrast with the corpus-internal upper bound is the frequency result. Corpus-level scores should accompany document, case and language results.

For TAB and OntoNotes, a scheme-aware adversary can derive the placeholder release from the surrogate release. Replacement form still affects utility. Telling the ranker the scheme raises recovery from 0.110 to 0.165 on unchanged text. In the tested tasks, surrogates preserve token-level performance and placeholders more document-level utility.

The prediction that sensitivity moves frequency alignment and linkage in opposite directions is refuted for these attacks, priors, corpora and ensemble. Better detection was expected to starve context linkage while enlarging the population available for frequency alignment. Frequency alignment recovered zero identities at every tested sensitivity with these priors, while both linkage attacks declined as sensitivity rose. In these experiments, missed identifiers limited both measured attacks.

Linkage is measurable only on Enron and CARDIO:DE because TAB and OntoNotes link mentions within documents. People recur there, so the reported linkage values understate rather than bound their risk. CARDIO:DE names were inserted into an already de-identified release, making its frequency distribution synthetic and its clinical realism partial. Dates pass through by design, leaving that channel open. The candidate-ranking attack presents ten names and therefore represents an adversary with a shortlist, not population search. Every number describes one release under one key.

5 Conclusion

Safe sharing of clinical data needs both modalities protected: a guarded image does not compensate for a report that names the patient, and a guarded report does not remove identity from the image. Artificial intelligence makes both possible at scale, but its failures must be measured where data are released. For text we priced detection, utility and re-identification on the same documents. A large union ensemble replaces almost every person mention and holds both attacks near their chance rates, and what remains is bounded by the identifiers the detector missed rather than by the replacement scheme, which is one rewrite away from its alternative. Detector precision governs utility more than the choice between surrogates and placeholders, and sensitivity understates exposure per document, case and language. There is therefore no single anonymisation score: a release should state its operating point, per-language errors, surviving cases, attacker knowledge and stability. Our operating points and measurements are released with it.

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